Name: ____________________________ Exit quiz

Generating a sequence using a term-to-term rule

1 Which of these best describes what a term-to-term rule is? Tick **1** correct answer

- ☐ A rule to find any term in a sequence
- ☐ A rule to find the first term in a sequence
- ☐ A rule to find the next term in a sequence from the previous term
- ☐ A rule to find the multiplier for a sequence
- ☐ A rule to find the constant additive pattern for a sequence

2 Match the term-to-term rules with the sequences Write the correct letter in each box

a	8, 12, 18, 27, ...
b	8, 12, 16, 20, ...
c	8, 12, 20, 32, ...
d	8, 12, 20, 36, ...
e	8, 12, 17, 23, ...

	multiply the previous term by $\frac{3}{2}$
	add 4, then add 5, then add 6, ...
	add 4 to the previous term
	add 4, then add 8, then add 12, ...
	double the previous term then subtract 4

3 A sequence starts 7,10, ... What is the 10th term if the term to term rule is 'add 3 to the previous term'? Fill in the blank

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4 Which of the statements below is a limitation of using term-to-term rules? Tick **3** correct answers

- ☐ They can be time consuming or complex to use when finding a larger term number.
- ☐ They can be difficult to use when you want to find the next term.
- ☐ Different sequences can be made from the same rule with different 1st terms.
- ☐ Different rules can apply to the same few terms.
- ☐ They only work for constant additive patterns.

5 What is the term-to-term rule for this sequence with a constant additive pattern? 17, ?, 29, ?, 41, ... Tick **1** correct answer

- [ ] add 6 to the previous term
- [ ] add 11 to the previous term
- [ ] add 12 to the previous term
- [ ] add 17 to the previous term
- [ ] add 24 to the previous term

6 What is the term-to-term rule for the constant additive sequence with 2nd term = 63 and 7th term = 23? Tick **1** correct answer

- [ ] add -10 each time
- [ ] add -8 each time
- [ ] add -5 each time
- [ ] add 8 each time
- [ ] add 40 each time